L1

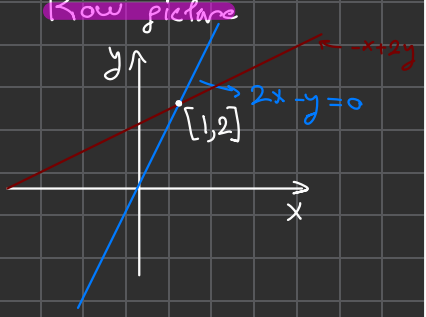
$2x - y = 0$

$-x + 2y = 3$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$A \quad x = b$

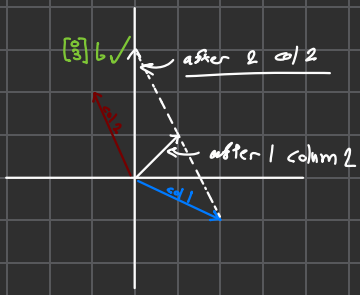
Row picture



Colum picture

$$x\begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad y\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

linear combination of columns



3. eq 3 unknown

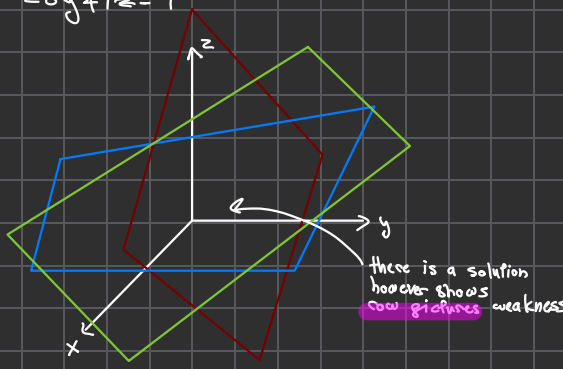
- $2x - y = 0$
- $-x + 2y - z = -1$
- $-3y + 4z = 4$

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & +2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

$A \quad x = b$

$$\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

$x \quad y \quad z \quad b$



column picture



how to multiply a matrix by x

$Ax = b$

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

$1 \cdot 2 + 2 \cdot 5 = 12$
$1 \cdot 1 + 2 \cdot 3 = 7$

$\begin{bmatrix} 12 \\ 7 \end{bmatrix}$

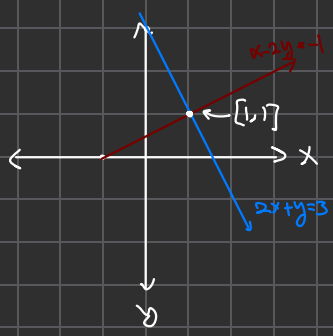
Ax is the combinations of the colums by A

Questions

Solve $\begin{bmatrix} 2x+y=3 \\ x-2y=-1 \end{bmatrix}$ and find out its row picture and colum picture

$$x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

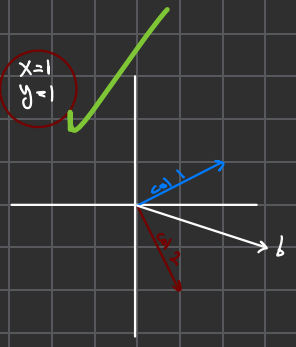
col 1 col 2 b



- $2x+y=3$
- $x-2y=-1$

$x=1 \quad y=1$

row view



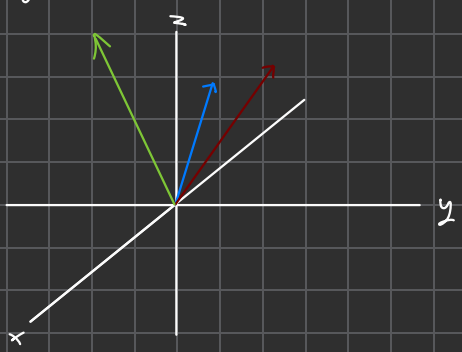
matrix form

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$A \quad x \quad = \quad b$

$A^{-1} \cdot A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

problem 1.1



$$x_1 + 4x_2 + 7x_3 = 0$$
$$2x_1 + 5x_2 + 8x_3 = 0$$
$$3x_1 + 6x_2 + 9x_3 = 0$$

$$x_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + x_2 \begin{bmatrix}4\\5\\6\end{bmatrix} + x_3 \begin{bmatrix}7\\8\\9\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

w_1 w_2 w_3 b

$3x_2 + 6x_3 = 0 \quad x_2 + 2x_3 = 0$

-2

$6x_2 + 12x_3 = 0$

$x_1 = 1 \quad x_2 = -2 \quad x_3 = 1$

problem 1.2

multiply $\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 3\\ 4 & 1 & 1\end{bmatrix}\begin{bmatrix}3\\-2\\1\end{bmatrix}$

$3 - 4 = -1$

$6 + 3 = 9$

$12 - 2 + 1 = 11$

$\begin{bmatrix}-1\\9\\11\end{bmatrix}$

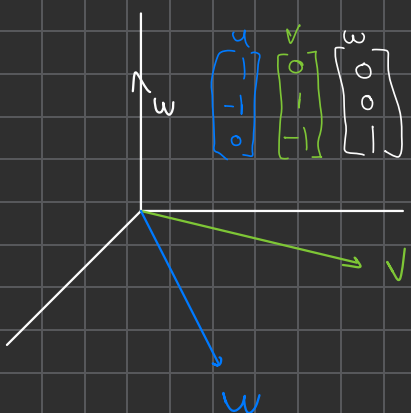
problem 1.3

true or false a 36 by 2 matrix A time 2 by 3 matrix b = 3.3 A6

L2

vectors/matrices/subspaces

any b $\quad x_1 u + x_2 v + x_3 w = b$ ← scalars



$$A_d \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix}$$

→ first difference matrix

$$A \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Solve for $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_1 + b_2 \\ b_1 + b_2 + b_3 \end{bmatrix}$$

$x_1 = b_1$

$x_2 - x_1 = b_2 \quad x_2 = b_1 + b_2$

$x_3 - x_2 = b_3 \quad x_3 - b_1 - b_2 = b_3 \quad x_3 = b_1 + b_2 + b_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

→ A^{-1}

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$Ax = b$

$x = bA^{-1}$

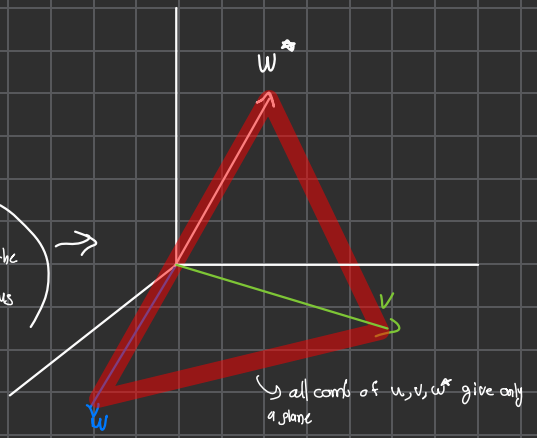
$$C\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}1&0&-1\\-1&1&0\\0&-1&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}x_1-x_3\\x_2-x_1\\x_3-x_2\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\b_3\end{bmatrix}$$

$\rightarrow 0 = b_1+b_2+b_3$

$$C = \begin{bmatrix}1&0&-1\\-1&1&0\\0&-1&1\end{bmatrix}\begin{bmatrix}c\\c\\c\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$Cx = 0$

they are in the same plane thus dependent



↳ all combs of u, v, w* give only a plane

= all vectors $C*$

then what $\begin{bmatrix}b_1\\b_2\\b_3\end{bmatrix}$ do we get?

$A_{3\times3}$ = uninvertable

L3

$$x+2y+z=2$$
$$3x+8y+z=12$$
$$4y+4z=2$$

elimination → success / fail → success / fail → success / ...

first pivot

$$\begin{bmatrix}1&2&1\\3&8&1\\0&4&1\end{bmatrix} \rightarrow \begin{bmatrix}1&2&1\\0&2&-2\\0&4&1\end{bmatrix} \rightarrow \begin{bmatrix}1&2&1\\0&2&-2\\0&0&5\end{bmatrix}$$

A → upper triangle U

back substitution

$$\left[\begin{array}{ccc|c}1&2&1&2\\3&8&1&12\\0&4&1&2\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&2&1&2\\0&2&-2&6\\0&4&1&2\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&2&1&2\\0&2&-2&6\\0&0&5&-10\end{array}\right]$$

A b U c

$x+2y+z=2$ $\quad z=-2 \quad y=1 \quad x=2$

$2y-2z=6$

$5z=-10$

matrices

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{matrix} 3\times \text{column 1} \\ 4\times \text{column 2} \\ 5\times \text{column 3} \end{matrix}$$

Matrix × column = column

row × matrix = row

$$\underset{1\times 3}{\begin{bmatrix} 1 & 2 & 7 \end{bmatrix}} \underset{3\times 3}{\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}} = \begin{matrix} 1\times \text{row one} \\ 2\times \text{row two} \\ 7\times \text{row 3} \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

E_{21}

subtract 3 row 1 from row 2

step two (subtract)

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

$E_{32}(E_{21}A) = U \qquad (E_{32}E_{21})A = U$

$$\begin{bmatrix}1&0&0\\0&1&0\\0&-2&1\end{bmatrix}\left(\begin{bmatrix}1&0&0\\-3&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&2&1\\3&8&1\\0&4&1\end{bmatrix}\right)=\begin{bmatrix}1&2&1\\0&2&-2\\0&0&5\end{bmatrix}$$

E_{32} E_{21} A U

$$\begin{bmatrix}-&-&-\\-&-&-\\-&-&-\end{bmatrix}\begin{bmatrix}1&2&1\\3&8&1\\0&4&1\end{bmatrix}=\begin{bmatrix}1&2&1\\0&2&-2\\0&0&5\end{bmatrix}$$

$(E_{32}E_{21})$ A U

permutation

exchange row 1 and 2

$$\underbrace{\begin{bmatrix}0&1\\1&0\end{bmatrix}}_{P}\begin{bmatrix}a&b\\c&d\end{bmatrix}=\begin{bmatrix}c&d\\a&b\end{bmatrix}$$

exchange column 1 and 2

$$\begin{bmatrix}a&b\\c&d\end{bmatrix}\underbrace{\begin{bmatrix}0&1\\1&0\end{bmatrix}}_{P}=\begin{bmatrix}b&a\\d&c\end{bmatrix}$$

row operations are on the left

column operation are on the right

inverses

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

E^{-1} E I

problems

solve using the method of elimination

$$\begin{aligned} x - y - z + w &= 0 \\ 2x \quad + 2z \quad &= 8 \\ -y - 2z \quad &= -8 \\ 3x - 3y - 2z + 4w &= 7 \end{aligned}$$

pivot

$$\begin{bmatrix} 1 & -1 & -1 & 1 \\ 2 & 0 & 2 & 0 \\ 0 & -1 & -2 & 0 \\ 3 & -3 & -2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 & 1 \\ 0 & 2 & 4 & -2 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 & 1 \\ 0 & 2 & 4 & -2 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & -1 & 1 & 0 \\ 0 & 2 & 4 & -2 & 8 \\ 0 & 0 & 1 & 1 & 7 \\ 0 & 0 & 0 & -1 & -4 \end{array}\right]$$

$w = 4$

$z + 4 = 7 \quad z = 3$

$2y + 12 - 8 = 8 \quad y = 2$

$x - 1 = 0 \quad x = 1$

$w = 4 \quad z = 3 \quad y = 2 \quad x = 1$

problem 2.1: 2.1.1 in the two-by-two system of linear equation below, what multiple of the first equation should be subtracted from the 2nd equation when using the method of elimination? 2.1.2 find the pivots. 2.1.3 find a solution.

$2x + 3y = 5$
$6x + 18y = 12$

given with question

$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} =$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

$\rightarrow \begin{bmatrix} 2 & 3 & 5 \\ 0 & 6 & -3 \end{bmatrix} \rightarrow$

$2x + 3y = 5$
$6y = -3$

$x = 3.25$
$y = -\frac{1}{2}$

Q 2.1.1

$-3 \begin{bmatrix} 2 & 3 \\ 6 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 12 \end{bmatrix}$

A b

Q 2.1.2 (these are the pivots)

Q 2.1.3 solutions

$2x = 6.5 \quad 3y = -1.5$
$6x = 19.5 \quad 15y = -7.5$

problem 2.2: 2.2.1 find the triangular matrix E that reduces "pascalls matrix" to a smaller pascl

$E \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

row operations

$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} =$

2.2.2 Which matrix M (multiplying several E's) reduces Pascal all the way to I?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

E_1 $\qquad$ A

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{-1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$M = E_3 \cdot E_2 \cdot E_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

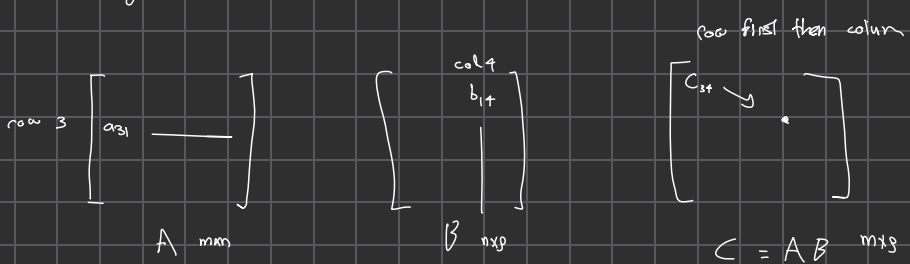
E_2 $\qquad$ E_1

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

E_3 $\qquad$ $E_2 E_1$

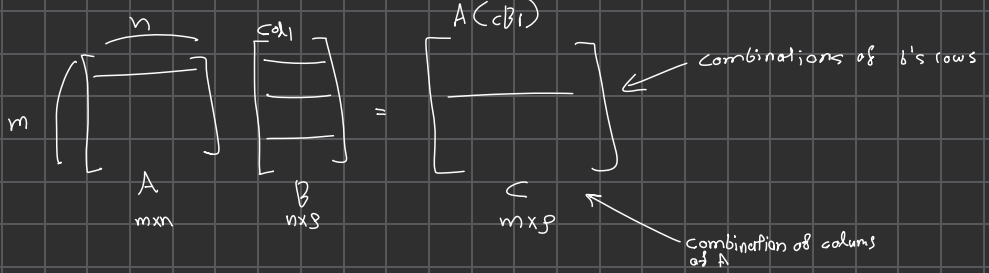
L4

4 ways of matrix multiplication and inverse of A $AB A^T$ find A^{-1}



C_{34} = (row 3 of A) . (column 4 of b)

$$C_{34} = a_{31} \cdot b_{14} + a_{32} \cdot b_{24} + a_{33} \cdot b_{34} + \cdots = \sum_{k=1}^{n} a_{3k}\, b_{k4}$$



Colum of A x row of B

m x 1 1 x p

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}$$

A B

$$\begin{bmatrix} 2 & 1 & 6 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = 31$$

4th way

AB = Sum of (col of A x rows of B)

$$\begin{matrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{matrix} \times \begin{matrix} 1 & 6 \\ 0 & 0 \end{matrix} = \begin{matrix} 2 & & 7 \\ 3 & 1 & 6 \\ 4 & & 9 \end{matrix} + \begin{matrix} 8 & 0 & 0 \end{matrix}$$

$$= \begin{vmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{vmatrix}$$

5th Blocks multiplication

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \times \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1B_1+A_2B_3 & \\ & \end{bmatrix}$$

A B

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{matrix} 19 & 22 \\ 43 & 50 \end{matrix}$$

1 of row 1 + 2 of row 2

3 of row 1 + 4 of row 2

Inverses (square matrices)

$A^{-1}A = I = AA^{-1}$ if this exists

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \times \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

3 of this

−1 of this

← singular

invertible, nonsingular

I can find a vector x with $Ax = 0$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A A^{-1}

Gauss-Jordan (solve 2 equations at once)

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right] \quad \begin{matrix} A & & I & \\ & & & \end{matrix}$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array}\right]$$

I A^{-1}

A x columns of A^{-1} = columns of I //

Find the condition on a and b that makes A invertible and find A^{-1}

$a \neq 0 \quad a \neq b$

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$$

$$[A|I] \xrightarrow[A^{-1}]{} [I|A^{-1}]$$

$$\begin{bmatrix} a & b & b & 1 & 0 & 0 \\ a & a & b & 0 & 1 & 0 \\ a & a & a & 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} a & b & b & 1 & 0 & 0 \\ 0 & a-b & 0 & -1 & 1 & 0 \\ 0 & a-b & a-b & -1 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} a & b & b & 1 & 0 & 0 \\ 0 & a-b & 0 & -1 & 1 & 0 \\ 0 & 0 & a-b & 0 & -1 & 1 \end{bmatrix}$$

$$\longrightarrow \begin{bmatrix} 1 & b/a & b/a & 1/a & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & b/a & b/a & 1/a & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{1}{a-b} & 0 & \frac{b}{ba-a^2} \\ 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{b-a} & \frac{1}{a-b} \end{bmatrix}$$

$(row1 - (row2 \times b/a + row3 \times b/a))$

So then we can say A^{-1} is

$$\begin{bmatrix} \frac{1}{a-b} & 0 & \frac{b}{ba-a^2} \\ \frac{1}{b-a} & \frac{1}{a-b} & 0 \\ 0 & \frac{1}{b-a} & \frac{1}{a-b} \end{bmatrix} = \frac{1}{a-b} \begin{bmatrix} 1 & 0 & -\frac{b}{a} \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$



Problem 3.1 Add AB to AC and compare with $A(B+C)$:

$$A=\begin{bmatrix}1&2\\3&4\end{bmatrix} \quad B=\begin{bmatrix}1&0\\0&0\end{bmatrix} \quad C=\begin{bmatrix}0&0\\5&6\end{bmatrix}$$

Solve:

$$AB=\begin{bmatrix}1&0\\3&0\end{bmatrix} \quad AC=\begin{bmatrix}10&12\\20&24\end{bmatrix} \quad AB+AC=\begin{bmatrix}11&12\\23&24\end{bmatrix}$$

$$B+C=\begin{bmatrix}1&0\\5&6\end{bmatrix} \quad A=\begin{bmatrix}1&2\\3&4\end{bmatrix} \quad A(B+C)=\begin{bmatrix}11&12\\23&24\end{bmatrix}$$

$$\rightarrow A(B+C)=AB+AC$$

✓

Problem 3.2 Use Gauss Jordan elimination on $[U,I]$ to find the upper triangular U^{-1}:

$$UU^{-1}=I \qquad \begin{bmatrix}1&a&b\\0&1&c\\0&0&1\end{bmatrix} \times \begin{bmatrix}x_1&x_2&x_3\\&&\end{bmatrix} = \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

U $\quad$ U^{-1} $\quad$ I

Solve:

$$\begin{bmatrix}1&a&b&1&0&0\\0&1&c&0&1&0\\0&0&1&0&0&0\end{bmatrix} \longrightarrow \begin{bmatrix}1&0&b-ac&1&-a&0\\0&1&0&0&1&-c\\0&0&1&0&0&1\end{bmatrix} \longrightarrow \begin{bmatrix}1&0&0&1&-a&ac-b\\0&1&0&0&1&-c\\0&0&1&0&0&1\end{bmatrix}$$

$-R_2\times a+R_1=R_1$ $\quad -cR_3+R_2=R_2$ $\quad -R_3(b-ac)+R_1=R_1$

Therefore $U^{-1}=\begin{bmatrix}1&-a&ac-b\\0&1&-c\\0&0&1\end{bmatrix}$

✓

15

$$(AB)(B^{-1}A^{-1}) = I \quad (I)$$

$$I^2 = I$$

$$B^{-1}A^{-1}AB = I$$

$$AA^{-1} = I$$

$$(A^{-1})^T(A)^T = I$$

$$(A^{-1})^T = (A^T)^{-1}$$

T and -1 can be done in either order for a single matrix

$$E_{21}\begin{bmatrix}1&0\\-4&1\end{bmatrix} \times A\begin{bmatrix}2&1\\8&7\end{bmatrix} = U\begin{bmatrix}2&1\\0&3\end{bmatrix}$$

$$A\begin{bmatrix}2&1\\8&7\end{bmatrix} = L\begin{bmatrix}1&0\\4&1\end{bmatrix} \times U\begin{bmatrix}2&1\\0&3\end{bmatrix}$$

$$\begin{bmatrix}1&0\\4&1\end{bmatrix}\begin{bmatrix}2&0\\0&3\end{bmatrix}\begin{bmatrix}1&\frac{1}{2}\\0&1\end{bmatrix}$$

$E_{32}E_{31}E_{21}A = U$ (no row exchanges)

$$A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U$$

$$A = \underbrace{L}_{\text{Lower triangle}}U$$

$$E_{32}\begin{bmatrix}1&0&0\\0&1&0\\0&-5&1\end{bmatrix} \quad E_{21}\begin{bmatrix}1&0&0\\-2&1&0\\0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0\\-2&1&0\\10&-5&1\end{bmatrix} = E$$

normal $\left(\begin{matrix}\text{left of}\\A\end{matrix}\right)$ $EA = U$

$$\begin{bmatrix}1&0&0\\2&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\0&5&1\end{bmatrix} = \begin{bmatrix}1&0&0\\2&1&0\\0&5&1\end{bmatrix} = L$$

inverse $\left(\begin{matrix}\text{left of}\\U\end{matrix}\right)$ $A = LU$

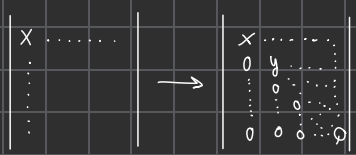
How many operations in an $n \times n$ matrix A?

$n = 100$

$n^2 + (n-1)^2 + (n-2)^2 \ldots 1^2$ = operation cost $\simeq \frac{1}{3}n^3$

$A = LU$

if no row exchanges, multipliers go directly into L



Permutations 3×3 6 P's $P^{-1} = P^T$

4×4 = 4! P's

$$\begin{bmatrix}1&&\\&1&\\&&1\end{bmatrix}, P_{12}\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$$

problems.

Find the LU-decomposition of matrix: / when it exists. / for which real numbers a and b does it exist?

$$A = \begin{vmatrix} 1 & 0 & 1 \\ a & a & a \\ b & b & a \end{vmatrix}$$

Solution:

so $EA = U$ and $A = LU$

so lets find U first.

$$\underbrace{\begin{vmatrix} 1 & 0 & 1 \\ a & a & a \\ b & b & a \end{vmatrix}}_{A} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}}_{E_{21}} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ b & b & a \end{vmatrix} \qquad \begin{vmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ b & b & a \end{vmatrix} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & 0 & 1 \end{vmatrix}}_{E_{31}} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & b & a-b \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & b & a-b \end{vmatrix} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{b}{a} & 1 \end{vmatrix}}_{E_{32}} = \underbrace{\begin{vmatrix} 1 & 0 & 1 \\ 0 & a & 0 \\ 0 & 0 & a-b \end{vmatrix}}_{U} \leftarrow \text{this is the } U \text{ matrix}$$

now lets find L:

$$A = LU = E_{21}^{-1} E_{31}^{-1} E_{32}^{-1} U$$

$$L = \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}}_{E_{21}} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & 0 & 1 \end{vmatrix}}_{E_{31}} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{b}{a} & 1 \end{vmatrix}}_{E_{32}} \qquad L = \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{vmatrix}}_{E_{21,31}} \times \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{b}{a} & 1 \end{vmatrix}}_{E_{32}} = \underbrace{\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & \frac{b}{a} & 1 \end{vmatrix}}_{L}$$

for this to exists only for $a \neq 0$

P4.1.1: what matrix E puts A into triangular form $EA = U$?

P4.1.2: Multiply by $E^{-1} = L$ to factor A into LU.

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

A

Solution: firstly lets solve for U:

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 2 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

A $\qquad$ E_{21}

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & 1 \end{pmatrix}$$

E_{31}

$$\begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

E_{32} $\qquad$ U



now for $A = LU$ lets find L where L is $E_{21}^{-1} E_{31}^{-1} E_{32}^{-1}$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$$

E_{21}^{-1} $\qquad$ E_{31}^{-1} $\qquad$ E_{32}^{-1} $\qquad$ L



now for $A = LU$ we get the matrices:

$$A = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 2 & 0 & 1 \end{vmatrix} = LU = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 3 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$



P 4.2.01: Compute L and U for the symmetric matrix and find four conditions for a, b, c, d

$$A = \begin{vmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{vmatrix}$$

Solution:

$$\begin{vmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{vmatrix} \longrightarrow \begin{vmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{vmatrix} \longrightarrow \begin{vmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-a \\ 0 & 0 & c-b & d-a \end{vmatrix} \longrightarrow \begin{vmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-a \\ 0 & 0 & 0 & d-c \end{vmatrix} = U$$

$$E_1 = \begin{vmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \quad E_2 = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{vmatrix} \quad E_3 = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{vmatrix}$$

now that are found U, lets solve for E^{-1} or L

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{vmatrix}$$

E_1 E_2 E_3

now solve for E^{-1}

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{array}\right]$$

L

The conditions are

$$a \neq 0,\ b-a \neq 0,\ c-b \neq 0,\ d-c \neq 0$$

L6

Permutations P: execute row exchanges

$$A = LU = \begin{bmatrix} 1 & 0 \\ \square & 1 \end{bmatrix} \begin{bmatrix} \square & \square \\ 0 & \square \end{bmatrix}$$

$PA = LU$ / any invertible A

elimination with row exchanges

P = identity matrix with reordered rows / $n!$ = number of row reorderings $P^{-1} = P^T$ $P^T P = I$

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \leftarrow R^T \qquad \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} \leftarrow R$$

Transpose $(A^T)_{ij} = A_{ji}$ Symmetric matrices $A^T = A$

$$\begin{bmatrix} 3 & 5 \\ 5 & 2 \end{bmatrix}^T = \begin{bmatrix} 3 & 5 \\ 5 & 2 \end{bmatrix}$$

$R^T R$ is always symmetric

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 7 \\ 11 & - & - \\ 7 & - & - \end{bmatrix}$$

why? Take the transpose $(R^T R)^T = R^T R$

take 1,3 and 1,3 · do $1 \cdot 1 + 3 \cdot 3 = 10$

then take 2,3 and 1,3 (it's the first row) do $2 \cdot 1 + 3 \cdot 3 = 11$

then take 4,1 and 1,3 · do $4 \cdot 1 + 3 \cdot 1 = 7$

then 1,3 and 2,3 (for recognized row) ....

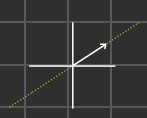
Vector spaces / sub spaces.

Examples: R^2 = all 2-dim. real vectors $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\begin{bmatrix} \pi \\ e \end{bmatrix}$ "x,y" plane / R^3 = all 3 dim. vectors $\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ / R^n = all (vectors with n components column)

Not a vector space:



A vector space inside R^2 (subspace):



to be a subspace the line in R^2 must go through the 0 vector.

Subspaces of R^2:

① all of R^2 ② any line through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ③ the zero vector

Subspaces of R^3:

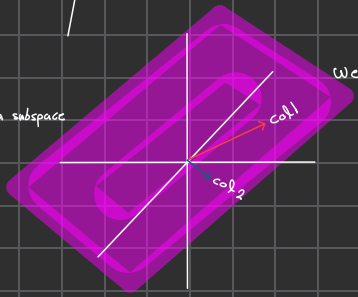
① all of R^3 ② plane ③ zero vector

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$$

columns are in R^3

all their linear combinations form a subspace

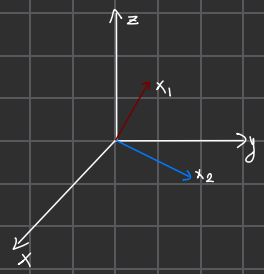
called the column space $C(A)$



we fill out a whole plane

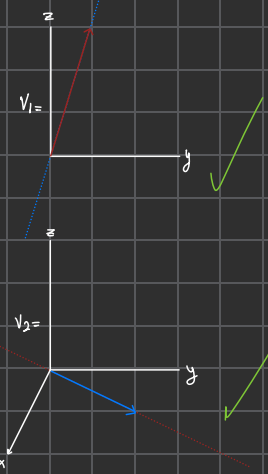
Problems.

$X_1 = \begin{vmatrix} 0 \\ 1 \\ 3 \end{vmatrix}$ $X_2 = \begin{vmatrix} 2 \\ 4 \\ 0 \end{vmatrix}$



1. Find V_1 = Subspace generated by X_1
   V_2 = Subspace generated by X_2
   Describe $V_1 \cap V_2$.

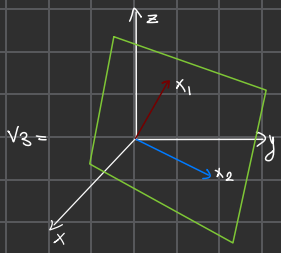
Solution:



$V_1 \cap V_2$ would be the $\begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$ vector

2. Find V_3 = Subspace generated by $[X_1, X_2]$
   is V_3 equal to $V_1 \cap V_2$?
   find subspace S of V_3 s.t $X_1 \notin S, X_2 \notin S$.

Solution:



V_3 would be a plane

no V_3 is a plane and wouldn't equal $V_1 \cap V_2$

S would be anything in the V_3 plane without x_1 and x_2

3. what is $V_3 \cap$ [xy plane] ?

Solution:

V_3 and xy plane would intersect at V_2

Problem 5.1:

a) Find a 3 by 3 permutation matrix with $P^3 = I$ $(P \neq I)$.

b) find a 4x4 permutation $\hat{P}$ where $\hat{P}^4 \neq I$.

Solution:

$$P \cdot P \cdot P = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$P = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

Solution:

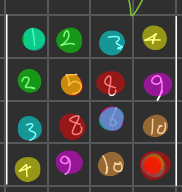
$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{vmatrix}$$

any one specific would work.

Problem 5.2 Suppose A is a four by four matrix, how many of A can be chosen independently if:

A) A is symmetric? 10

Solution



randomly chosen symmetric 4x4

B) A is skew-symmetric? $(A^T = -A)$?

since the symetrices would be opposite of each other we would get 16. ✗

it would be 6 because

$$\begin{matrix} 0 & -6 & -4 & -3 \\ 6 & 0 & -5 & -2 \\ 4 & 5 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{matrix}$$

Problem 5.3 True or False

a) symmetric matrices in M form a subspace. $(A^T = A)$

$$(A+B)^T = A^T + B^T = A + B \qquad (cA)^T = cA$$

b) skew-symmetric matrices in M form a subspace. $(A^T = -A)$ $(cA)^T = cA$

$$(A+B)^T = A^T + B^T = -A - B = -(A+B) \qquad (cA)^T = -cA$$

c) unsymmetric matrices for a subspace in $M (A^T \neq A)$

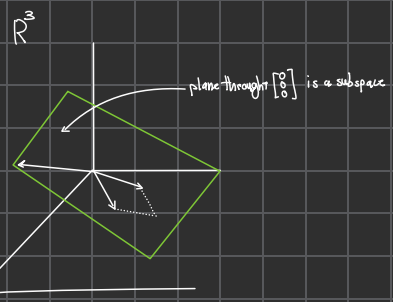
$$\begin{vmatrix} 3 & 0 \\ 1 & 4 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ 0 & 4 \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix}$$

L7

vector space requirements

$v+w$ and cv are in the space

$cv+dw$ are in as well



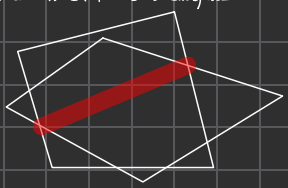
2 subspaces: P and L

P ∪ L = all vectors in P or L or both) this is not a subspace because i can go outside of the union

P ∩ L = all vector in both P and L

Subspace S and T

intersection S ∩ T is a subspace



Column space of A is a subspace of $R^?$ $C(A)$

$$A=\begin{bmatrix}1&1&2\\2&1&3\\3&1&4\\4&1&5\end{bmatrix}$$

so $C(A)$ = all linear combinations of columns

Does $Ax=b$ always have a solution for every b? no

4 equations, 3 unknowns

$$Ax=\begin{bmatrix}1&1&2\\2&1&3\\3&1&4\\4&1&5\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}b_1\\b_2\\b_3\\b_4\end{bmatrix}$$

which b's allow this system to be solved?

$$\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$$

so for any x_1, x_2, x_3 (when b is in the column space)

A x b

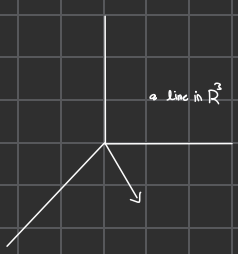
Nullspace of A = all solutions $x=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$ to $Ax=0$ $R^?$

$N(A)$ contains $k\begin{bmatrix}1\\1\\1\end{bmatrix}$

$$\begin{bmatrix}1&1&2\\2&1&3\\3&1&4\\4&1&5\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

A x

$N(A)$ =



check that solution to $Ax=0$ are always in a subspace.

so $Av=0$ and $Aw=0$ then $A(v+w)=0=Av+Aw$

then $A(cv)=0$

Problems:

Which are subspaces of R^3 $\begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix}$

1) $b_1 + b_3 - b_3 = 0 \rightarrow \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \times \begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix} = 0$ this is a subspace of R^3 ✓

2) $b_1\ b_2 - b_3 = 0$ this isn't a subspace of R^3 ✓

3) $\begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \\ 0 \end{vmatrix} + c_1 \begin{vmatrix} 1 \\ 0 \\ -1 \end{vmatrix} + c_2 \begin{vmatrix} 1 \\ 0 \\ 1 \end{vmatrix} \rightarrow \begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix} = \begin{vmatrix} 1 \\ c_1 + \frac{1}{2} \\ -1 \end{vmatrix} + c_1 + \frac{1}{2} \begin{vmatrix} 1 \\ 0 \\ 1 \end{vmatrix}$ this is a subspace of R^3 ✓

4) $\begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix} = \begin{vmatrix} 0 \\ 1 \\ 0 \end{vmatrix} + c_1 \begin{vmatrix} 1 \\ 0 \\ -1 \end{vmatrix} + c_2 \begin{vmatrix} 1 \\ 0 \\ 1 \end{vmatrix}$ this isn't a subspace of R^3 ✓

Problem 6.1: Suppose S and T are two subspaces of a vector space V

The sum S+T contains all sums s+t of a vector s in S and t in T. Show that S+T is a vector space

$v_1 = s_1 + t_1 \quad v_2 = s_2 + t_2 \quad v_1 + v_2 = s_1 + s_2 + t_1 + t_2 = k$

$k \cap (S \cup T) \neq \emptyset$ ✓

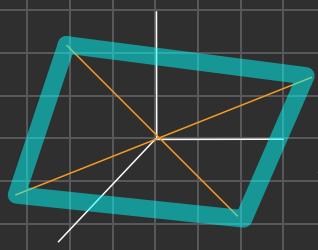
If S and T are lines in R^m, what's the difference between S+T and S∪T?

Explain "the span of S∪T is S+T": if we take all linear combination of S∪T we get S+T ✓

Lets say m is 3



✓



✓

S∪T (not a subspace since $x_1 + x_2 = x_3$ and $(S \cup T) \cap x_3 = \emptyset$)

S+T on the other hand creates a plain

Problem 6.2: the plane $x-3y-z=12$ is parallel to plane $x-3y-z=0$. One particular point on this plane is $(12,0,0)$. all points on the plane have the form.

(fill in the components)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad x = 12 + 3y + z$$ ✓

Problem 6.3: How is the null space $N(C)$ related to $N(A)$ and $N(B)$ if $C = \begin{bmatrix} A \\ B \end{bmatrix}$

ok lets say

$A = \begin{pmatrix} 2 & 1 \end{pmatrix}$ $\quad B = \begin{pmatrix} 3 & 1 \end{pmatrix}$ $\quad \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} = C$ so for $\begin{bmatrix} 2 & 1 & x_1 \\ 3 & 1 & x_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ so $Cx = \begin{bmatrix} Ax \\ Bx \end{bmatrix}$ so $(Ax=0 \cap Bx=0) = N(C)$ ✓

L8

Solve for $Ax=0$

The Rank of A = number of pivots = 2

$$A = \begin{bmatrix} \textcircled{1} & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 2 & 2 \\ 0 & 0 & \textcircled{2} & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 2 & 2 \\ 0 & 0 & \textcircled{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$x_1 + 2x_2 + 2x_3 + 2x_4 = 0$

$2x_3 + 4x_4 = 0$

$r = 2$

$n - r = 4 - 2 =$ Free variables

2 pivot columns

2 Free columns

$$x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

R = Reduced row echelon form (zeros above and bellow the pivots)

notice I in the pivot rows/columns $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ pivot part I $\qquad \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$ Free Part

$x_1 + 2x_2 - 2x_4 = 0$

$x_3 + 2x_4 = 0$

$$x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

notice I in the pivot rows/columns

$$\left[\begin{array}{cccc} \textcircled{1} & 2 & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} \qquad \begin{matrix} 2 & -2 \\ 0 & 2 \end{matrix}$$

pivot part I — Free Part

$$x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$x_1 + 2x_2 - 2x_4 = 0$

$x_3 + 2x_4 = 0$

So why did that happen, Lets assume we have this matrix in rref form

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

r pivot rows; n−r free columns

r pivot columns

$Rx = 0$ — null space matrix (columns = special solutions)

$$RN = 0 \qquad N = \begin{bmatrix} -F \\ I \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 3 \\ 0 & \textcircled{2} & 2 \\ 0 & 0 & 0 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{2} & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{c} \begin{matrix} I & & F \end{matrix} \\ \begin{bmatrix} \boxed{1} & 0 & 1 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

↑ ↑ u — ↘ Free column

$$x = c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = c \begin{bmatrix} -F \\ I \end{bmatrix}$$

↑ N

Problem set

(Fill in the blanks) The set S of points $p(x,y,z)$ s.t $x-5y+2z=9$ is a plain in $\vec{R}$. it is parallel to the plain of $p(x,y,z)$ - St. $x-5y+2z=0$

All points of s have the form — some point is $x-5y+2z=0$

$$\begin{matrix} x \\ y \\ z \end{matrix} = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

P_0 is in S $x-5y+2z=9$

Problem 7.1:

a) find the row reduced form of Matrix A:

$$\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 5 & 7 & 9 \\ 0 & \textcircled{4} & 1 & 7 \\ 0 & -12 & -3 & -21 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 5 & 7 & 9 \\ 0 & \boxed{4} & 1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{23}{4} & \frac{1}{4} \\ 0 & 1 & \frac{1}{4} & \frac{7}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b) what is the rank of this matrix? 2, because it has 2 pivots.

c) find any special solutions to $Ax=0$.

$$\begin{bmatrix} \textcircled{1} & 0 & \frac{23}{4} & \frac{1}{4} \\ 0 & \textcircled{1} & \frac{1}{4} & \frac{7}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad x = c \begin{bmatrix} -\frac{1}{4} \\ -\frac{7}{4} \\ 0 \\ 1 \end{bmatrix} = k \begin{bmatrix} -\frac{23}{4} \\ -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}$$

$x_1 \quad \frac{23x_3}{4} \quad \frac{x_4}{4}$

$x_2 \quad \frac{x_3}{4} \quad \frac{7x_4}{4}$

Problem 7.2:

Find A_1 and A_2 so that $\text{rank}(A_1, B) = 1$ and $\text{rank}(A_2 B) = 0$ for $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$A_1 B_1 = \begin{bmatrix} \boxed{1} & 1 \\ 1 & 1 \end{bmatrix} \qquad A_2 B_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

L9

$Ax=b$ if it has a solution, determen using elimination, if one exists then how many? find them all

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} \boxed{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \boxed{2} & 4 & b_2-b_1 \\ 0 & 0 & 0 & 0 & b_2-b_3-b_1 \end{array}\right] \quad 0 = b_3 - b_2 - b_1$$

Solvability (condition on right hand sides)

$Ax=b$ solvable when b is in the column spaces of A ($C(A)$)

is a combination of cow of A gives the 0 row the same combination of b must give 0

augmented matrix $=[A\,b]$

To find complete sol'n to $Ax=b$

Step one: $x_{particular}$ = set all free variables to 0 the solve $Ax=b$ for the pivot variables here it would be $(x_2=0, x_4=0)$

$$\left[\begin{array}{cccc|c} \boxed{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \boxed{2} & 4 & b_2-b_1 \\ 0 & 0 & 0 & 0 & b_2-b_3-b_1 \end{array}\right]$$

$x_1 + 2x_3 = 1 \qquad x_1 = -2$

$2x_3 = 3 \qquad x_3 = 3/2$

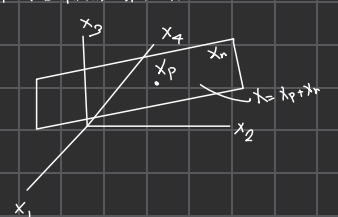
$$x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

Step two: x_n nullspace

Step three: $x_p + x_n$ = sol'n

$$\frac{\begin{array}{l} Ax_p = b \\ + \; Ax_n = 0 \end{array}}{A(x_p+x_n) = b}$$

plot all solutions x in R^4



$$x_{complete} = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

m by n matrix A of rank r (know column $r \le m$, $r \le n$)

full column Rank $r=n$: no free variables

$N(A)$ = only zero vector Solutions to $Ax=b$: $x = x_p$ (if there is one at all)

full column rank $r=n$

$$A = \left[\begin{array}{cc} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 8 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{array}\right]$$

0 or 1 solutions

full row rank $r=m$ Can solve $Ax=b$ for which right hand sides? for every b. Exists

Left with $\underset{n-m}{n-r}$ free variables.

here is an example of form

$A = \begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix}$ $R = \begin{bmatrix} 1 & 0 & -\frac{1}{5} & -\frac{7}{5} \\ 0 & 1 & -\frac{3}{5} & -\frac{4}{5} \end{bmatrix}$

$r = m = n$ full rank → invertable matrix

$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ $R = I$

$r=m=n$	$r=n<m$	$r=m<n$	$r<m$, $r<n$
$R=I$	$R=\begin{bmatrix} I \\ 0 \end{bmatrix}$	$R=[I\ F]$	$R=\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
1 solution	0 or 1 solution	always a solution	0 or ∞ solution
		∞ solutions	

problem set

Find all solutions, depending on b_1, b_2, b_3

$x-2y-2z=b_1$
$2x-5y-4z=b_2$
$4x-9y-8z=b_3$

$$\begin{bmatrix} 1 & -2 & -2 & b_1 \\ 2 & -5 & -4 & b_2 \\ 4 & -9 & -8 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 & b_1 \\ 0 & -1 & 0 & b_2-2b_1 \\ 0 & -1 & 0 & b_3-4b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 & b_1 \\ 0 & -1 & 0 & -2b_1+b_2 \\ 0 & 0 & 0 & -2b_1-b_2+b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 & 5b_1-2b_2 \\ 0 & 1 & 0 & 2b_1-b_2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

pivot, free

$b_2+b_3-2b_1=0$

$x_p\{Ax=b\} \Rightarrow z=0$ $x_p = \begin{bmatrix} 5b_1-2b_2 \\ 2b_1-b_2 \\ 0 \end{bmatrix}$ $x_n\{Ax=0\} \Rightarrow z=1$ $x_n = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$ $x_{complete} = x_p + C \cdot x_n$

problem 8.1: explain why these are false

A) The complete solution is any linear combination of x_p and x_n: False, because x_p is a singular solution so $x_{complete} = x_p + C_1 x_{n_1} + C_2 x_{n_2} \cdots C_n x_{n_n}$

B) The systems $Ax=b$ has all and one particular sol'un: False, what if $r=m<n$

C) If A is invertible there is no solution x_n in the nullspace: False because then $r=m=n$ there is always $x_n=0$

Problem 8.2 Let,

$$A=\begin{vmatrix}1&2&3\\0&0&4\end{vmatrix} \quad \text{und} \quad C=\begin{vmatrix}5\\8\end{vmatrix}$$

solution:

$$\left|\begin{array}{cccc}1&2&3&5\\0&0&4&8\end{array}\right| \rightarrow \left|\begin{array}{cccc}1&2&3&5\\0&0&\textcircled{1}&2\end{array}\right| \rightarrow \left|\begin{array}{cccc}1&2&0&-1\\0&0&\textcircled{1}&2\end{array}\right| = [R\ d]$$

$$\left|\begin{array}{cccc}1&2&3&0\\0&0&4&0\end{array}\right| \rightarrow \left|\begin{array}{cccc}1&2&0&0\\0&0&\textcircled{1}&0\end{array}\right| = [R\ 0]$$

$$R=\begin{vmatrix}1&2&0\\0&0&\textcircled{1}\end{vmatrix} \qquad R\cdot x=0=\begin{vmatrix}1&2&0\\0&0&\textcircled{1}\end{vmatrix}\begin{vmatrix}x_1\\x_2\\x_3\end{vmatrix}=\begin{vmatrix}0\\0\end{vmatrix}$$

$x_1+2x_2=0$

$x_3=0$

$$x=\begin{vmatrix}2\\-1\\0\end{vmatrix}$$

$$\left|\begin{array}{cccc}1&2&0&-1\\0&0&\textcircled{1}&2\end{array}\right| = [R\ d]$$

Since $x_p = R_K = d$

$$\left|\begin{array}{ccc}1&2&0\\0&0&\textcircled{1}\end{array}\right|\begin{vmatrix}k_1\\k_2\\k_3\end{vmatrix}=\begin{vmatrix}-1\\2\end{vmatrix}$$

$k_1+2k_2=-1$

$k_3=1$

$$k=\begin{vmatrix}-3\\1\\2\end{vmatrix}$$

$$\begin{vmatrix}1&2&3\\0&0&4\end{vmatrix}\cdot\begin{vmatrix}2\\-1\\0\end{vmatrix}=\begin{vmatrix}0\\0\end{vmatrix} \qquad \begin{vmatrix}1&2&3\\0&0&4\end{vmatrix}\cdot\begin{vmatrix}-3\\1\\2\end{vmatrix}=\begin{vmatrix}5\\8\end{vmatrix}$$

problem 8.3:

Suppose $Ax=b$ and $Cx=b$ have same solutions for b is it true that $A=C$?

Answer: yes because $x_{complete} = x_{particular} + x_{nullspace}$ if they have the same solutions then they are the same nullspace ✓

Although i made the right point I believe i should've explained further

L10

Suppose A is $m<n$ than there are nonzero solutions to $Ax=0$

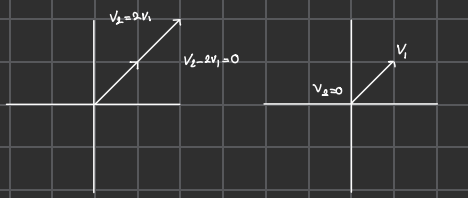
more unknown than equations)

Reason: there will be atleast one free variable!

atleast $n-m$ free variables

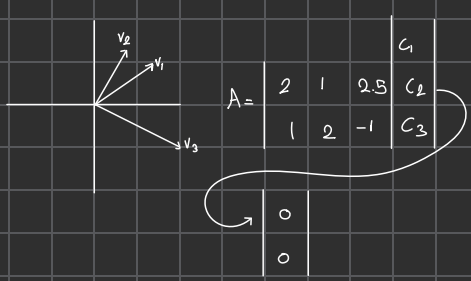
Independence: Vectors $x_1, x_2 \dots x_n$ are independent if no combination give the 0 vector (except $c_x=0$)

$c_1x_1 + c_2x_2 + \dots + c_nx_n = 0$



$0v_1 + xv_2 = 0$

$x=R$



Repeat when $v_1, \dots, v_n$ are columns of A.

independent if $N(A)$ is the [zero vector] • rank $= n$

dependent if not only the [zero vector] rank $< n$

a basis for a space is a sequence of vectors $v_1, v_2 \dots v_d$

with 2 properties

1- they are independent 2- they span the space

Example in R^3:

one basis is $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ another basis is $\begin{bmatrix}2\\1\\4\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}, \begin{bmatrix}1\\1\\3\end{bmatrix}$ any 3 independent R^3 vectors is a basis for R^3

culmy about the $N(A)$

$$\begin{bmatrix}1&2&3&1\\1&1&2&1\\1&2&3&1\end{bmatrix} \quad \begin{bmatrix}-1\\-1\\1\\0\end{bmatrix} \quad / \quad \begin{bmatrix}-1\\0\\0\\1\end{bmatrix}$$

dimension $N(A)$ = free variable

$N-r$

Space is $C(A)$ Rank (A) = #pivot columns = Dimension of $C(A)$

$$\begin{bmatrix}1&2&3&1\\1&1&2&1\\1&2&3&1\end{bmatrix}$$

not independent thus not a basis

but are a space

Problems:

Find the dimension of the vector space and find a basis for the space

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 1 & 2 & 0 & -4 & 1 \\ 0 & 1 & 3 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 1 & 3 & 3 & 0 \\ 0 & 1 & 4 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 1 & -2 & 0 & -1 \\ 0 & \textcircled{1} & 2 & -4 & 2 \\ 0 & 0 & \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \leftarrow \\ \leftarrow \\ \leftarrow \\ \end{matrix}$$

$R_2 = R_2 - R_1$ $\quad R_3 = R_3 - R_2$ $\quad R_4 = R_4 - R_3$

$R_4 = R_4 - 2R_1$ $\quad R_4 = R_4 - R_2$

The dimension is R^3 and the basis is: $\begin{vmatrix} \textcircled{1}, -2, 0, -1 \\ 0, \textcircled{1}, 2, -4, 2 \\ 0, 0, \textcircled{1}, 1, 0 \end{vmatrix}$

Problem 9.1: Find the largest possible number of independent vectors among

$v_1, v_2, v_3, v_4, v_5, v_6$

are the independent vectors

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 1 & 1 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 1 & 1 & 1 & 0 \\ 0 & 0 & \textcircled{1} & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 9.2: Find a basis for $X - 2y + 3z = 0$ in R^3 then find a basis for the intersection of that with the xy plane. Then find a basis for all vectors perpendicular to the plane

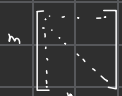
$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad Ax = 0 \quad x = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

v_1

$v_2 = \begin{matrix} -2 \\ 3 \end{matrix}$

L10

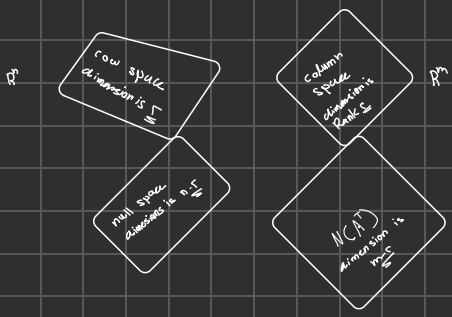
now being dependent means columns are dependent as well



Subspaces

Column space $C(A)$, Null space $N(A)$, row space = all combinations of rows of A = all combinations of the columns A^T so = $C(A^T)$, Null space of A^T = $N(A^T)$

R^m R^n R^r R^m



	$C(A)$	$N(A)$
basis?	pivot columns	special solutions
dimension?	r	$n-r$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

I F

$C(A) \neq C(R)$ different column spaces

Basis for row space is first r rows of R | Same row space

4th space: $N(A^T)$

$A^T y = 0$

$y^T A = 0^T$

$$\begin{bmatrix} & \end{bmatrix}\begin{bmatrix} \\ \end{bmatrix} = \begin{bmatrix} 0 \\ \end{bmatrix} \qquad \begin{bmatrix} y^T \end{bmatrix}\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

rref $[A \;\; I] \rightarrow [R \;\; E]$

$m \times n$ $m \times m$ $m \times n$ $m \times m$

$EA = R$ in chapter 2 R was I then E was A^{-1}.

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

I F

Next I would become $\begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = E$

new vector space! M A+B, cA (not AB for now)

all 3x3 matrices!!

subspaces of M	upper triangular	symmetric matrices	diagonal matrices
			D

dimension of this subspace is 3

problems

suppose

$$B = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$r = 2$

find a basis for and compute the dimension of each 4 fundm. subspace

C(B):

dimension is 2 (r)

basis: $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ ✓

N(B):

dimension is 1 (n-r)

a basis for N(B) is $\begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}$ ✓

C(B^T):

dimension is 2 (r)

basis for C(B^T): $\begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ ✓

N(B^T):

dimension is 1

basis for N(B^T): $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ ✓

$EB = U$

$$\begin{pmatrix} 1 & & \\ -2 & 1 & \\ 1 & 0 & 1 \end{pmatrix} \begin{bmatrix} 5 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Problem 10.1: A is an m by n matrix of rank r. suppose there are certain b's in which $Ax = b$ has no solution

what are the inequalities that must be true between m, n and r?

$Ax = b$ $r \leq n$ $r < m$ ✓

How do you know $A^T y = 0$ has other solutions than $y = 0$?

there has to be a left null space so $m - r > 0$ ✓

problem 10.2: $A^T y = d$ is solvable when d is in which of the four subspaces? and when is it unique?

when it is in the row space AKA $C(A^T)$

it can be unique when independent AKA the $N(A^T)$ has only the zero vector

L11

M = all 3 by 3 matrices dim = 9 S = Symmetric dim = 6 upper triangular dim = 6

Basis for M

$$\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&1&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix}, \ldots\ldots, \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}$$

$S \cap U$ = Symm and upper triangular = diagnol 3x3 $\dim(S \cap U) = 3$

$S + U = M$ $\dim(S+U) = 9$

$(\dim S = 6) + (\dim U = 6) = \dim(S+U) + \dim(S \cap U)$

$\frac{d^2y}{dx^2} + y = 0$ $y = \underbrace{\cos x, \sin x}_{\text{Basis}} \rightarrow y = c_1 \cos x + c_2 \sin x$ $\dim$ (solution space) = 2

$A = \begin{bmatrix}1&4&5\\2&8&10\end{bmatrix}$ 2x3 $r = 1$ a basis for A's row space is row 1 $\dim C(A) = \text{rank} = \dim C(A^T)$ the dimension of the column space is 1

$A = \begin{bmatrix}1\\2\end{bmatrix} \times [1\ 4\ 5]$ rank 1 matrices = $A = uv^T$

$r(A+B) \leq r(A) + r(B)$

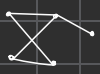
in $\mathbb{R}^4$ $V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$ S = all vectors V in $\mathbb{R}^4$ whir $v_1 + v_2 \dots v_4 = 0$ $AV = 0$ S = null space of A, $A = [1\ 1\ 1\ 1]$ $r = 1$ $\dim N(A) = n - r = 3$

basis for S

$\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ $3 \times 1 = 4 = n$ $1 \times 0 = 1 = m$

$C(A) = R_1$ $\quad N(A^T) = \{0\}$

Graph = nodes, edges



problem sets:

Show that the set of 2×3 matrixes whose null space contains $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ is a vector subspace, and find a basis for it.

$Ax = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$A \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $B \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad (CA) \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓

$(A+B) \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = A \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + B \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓ $\quad C\left(A \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓

$\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)$

$2x_1 + x_2 + x_3 = 0$

$x_1 \quad x_2 \quad -2x_1 - x_2$

$c_1 [1\ 0\ -2], c_2 [0\ 1\ -1]$

basis $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \end{bmatrix}$ ✓

problem 11.1

$P_{21} = \begin{bmatrix} 1 & & \\ 1 & & \\ & & 1 \end{bmatrix}$, $P_{31} = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix}$, $P_{32} = \begin{bmatrix} 1 & & \\ & & 1 \\ & 1 & \end{bmatrix}$, $P_{32} P_{21} = \begin{bmatrix} & 1 & \\ & & 1 \\ 1 & & \end{bmatrix}$

$P_{21} P_{32} = \begin{bmatrix} 1 & & \\ & & 1 \\ & 1 & \end{bmatrix}$ $\quad$ then $I = P_{21} + P_{31} + P_{32} - P_{32} P_{21} - P_{21} P_{32}$

for the second part $C_1P_1 + \cdots + C_5P_5 = 0$

$$\begin{vmatrix} c_3 & c_1+c_4 & c_2+c_5 \\ c_1+c_5 & c_2 & c_3+c_4 \\ c_2+c_4 & c_3+c_5 & c_1 \end{vmatrix} = 0$$

problem 112: M is the space of 3x3 matrices. Multiply each matrix x in M by:

$$A = \begin{vmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$ and notice that $A\begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$

A) which matrix x Leads to $Ax=0$

$$x = \begin{vmatrix} x_1 & x_2 & x_3 \\ x_1 & x_2 & x_3 \\ x_1 & x_2 & x_3 \end{vmatrix}$$ ✓

B) which matrices have the form Ax for some matrix x?

$Ax = k$

$$\therefore k = \begin{vmatrix} k_1 & k_2 & k_3 \\ k_4 & k_5 & k_6 \\ -k_1-k_4 & -k_2-k_5 & -k_3-k_6 \end{vmatrix}$$ ✓

C) part A find the Nullspace part b the column space what are their dimensions? and why do they add up to (n-c)+c=9?

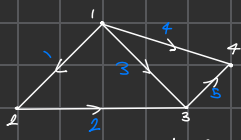
the null spaces dimension is 3 (3x1) ✓

while the column space is 6 (3x2) ✓

which add up to 9 which is the input dimension of M. (3x3) ✓

L13

graph: Nodes, edges



n = 4 nodes (column)

m = 5 edges (rows)

Incidence matrix

	node	1	2	3	4	
		-1	1	0	0	edge 1
		0	-1	1	0	2
	A =	-1	0	1	0	3
		-1	0	0	1	4
		0	0	-1	1	5

Loop

$A^T y = 0$ Kirchoff's CL

Solve $Ax = 0$ $x = x_1, x_2, x_3, x_4$ potentials of the nodes $\xrightarrow{C}$ Ohms Law on edges $\rightarrow$ currents $y_1\ y_2\ y_3\ y_4\ y_5$ (A^T)

$\xleftarrow{A}$ the differences in potential across the edges

$$Ax = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix} \qquad x = c \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

nothing will move if all are the same.

$e - Ax \quad y = Cx \quad A^T y = 0$ if we had current sources w edges: $A^T y = f$

$NCA = I$

$A^T y = 0$ $\dim(N(A^T)) = m - r = 5 - 3 = 2$

A^T is $n \times m$ 4×5

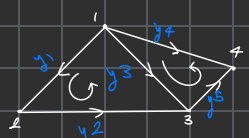
$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$-y_1 - y_3 - y_4 = 0$ node 1, net = 0

$y_1 - y_2 = 0$ node 2, net = 0

$y_2 + y_3 - y_5 = 0$ node 3, net = 0

$y_4 + y_5 = 0$ node 4, net = 0



Basis for $N(A^T)$ now for row space of A we do $C(A^T)$ and the columns 1, 2, 4 are independent and they are y_1, y_2, y_4 edges, they have no loop and this produces a tree (a graph with no loops)

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ -1 & 1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ \\ \\ \end{matrix} \begin{matrix} \\ \\ \\ \\ \\ \end{matrix}$$

$y_1 \quad y_2$

$\dim N(A^T) = m - r$ # nodes − # edges + # loops = 1 Euler's formula

loops = # edges − (number of nodes − 1)

(rank = n − 1)

problem set



find the incidence matrix A.

solution:

$$A = \begin{array}{c|ccccc|c} & \text{node } 2 & 3 & 4 & 5 & \\ \hline & -1 & +1 & 0 & 0 & 0 & \text{edges} \\ & 0 & -1 & +1 & 0 & 0 & 1 \\ & -1 & 0 & +1 & 0 & 0 & 2 \\ & 0 & -1 & 0 & +1 & 0 & 3 \\ & 0 & 0 & 0 & -1 & +1 & 4 \\ & 0 & 0 & +1 & 0 & -1 & 5 \end{array}$$

$N(A), N(A^T) = ?$

$Ax = 0 \quad N(A) = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ (the nodes have to have equal currents)

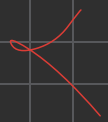
$A^T y = 0$

$$A^T = \begin{array}{|ccccc|} \hline -1 & 0 & -1 & 0 & 0 & 0 \\ +1 & -1 & 0 & -1 & 0 & 0 \\ 0 & +1 & +1 & 0 & 0 & +1 \\ 0 & 0 & 0 & +1 & -1 & 0 \\ 0 & 0 & 0 & 0 & +1 & -1 \\ \hline \end{array} \qquad y_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad y_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Trace (A^TA)=? (was not explained in the lecture)

degree of a node: the number of edges that connect to said node

trace: the sum of the degrees of the graph in the picture



$Tr(A^TA) = 2+3+3+2+2 = 12$

problem 12.1 write down the 4x4 incidence matrix A for the square graph, what vectors are in the nullspace of A? how do you know 1,0,0,0 is in the row space of A?



solutions:

$$A = \begin{array}{|cccc|l} \multicolumn{4}{c}{\text{nodes } 2 \quad 3 \quad 4} & \\ -1 & +1 & 0 & 0 & \text{edges} \\ 0 & -1 & +1 & 0 & 2 \\ 0 & 0 & +1 & -1 & 3 \\ -1 & 0 & 0 & +1 & 4 \end{array}$$

null space of $A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

because the current needs to balance

$A^Ty = 0$

problem 12.2: Suppose the conductance matrix find A^TCA. for $f = (1,0,-1,0)$, find a solution to $A^TCAx = f$. write the potencials x and currents $y = -CAx$ on the square graph for this current going into node 1 and out of node 3.

$$C = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad A^T = \begin{vmatrix} -1 & 0 & 0 & -1 \\ +1 & -1 & 0 & 0 \\ 0 & +1 & +1 & 0 \\ 0 & 0 & -1 & +1 \end{vmatrix} \quad A = \begin{vmatrix} -1 & +1 & 0 & 0 \\ 0 & -1 & +1 & 0 \\ 0 & 0 & +1 & -1 \\ -1 & 0 & 0 & +1 \end{vmatrix}$$

$$A^T(CA) = \begin{vmatrix} -1 & 0 & 0 & -1 \\ +1 & -1 & 0 & 0 \\ 0 & +1 & +1 & 0 \\ 0 & 0 & -1 & +1 \end{vmatrix} \times \left(\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} -1 & +1 & 0 & 0 \\ 0 & -1 & +1 & 0 \\ 0 & 0 & +1 & -1 \\ -1 & 0 & 0 & +1 \end{vmatrix} \right)$$

$$A^T(CA) = \begin{vmatrix} -1 & 0 & 0 & -1 \\ +1 & -1 & 0 & 0 \\ 0 & +1 & +1 & 0 \\ 0 & 0 & -1 & +1 \end{vmatrix} \times \begin{vmatrix} -1 & +1 & 0 & 0 \\ 0 & -2 & +2 & 0 \\ 0 & 0 & +2 & -2 \\ -1 & 0 & 0 & +1 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 4 & -2 \\ -1 & 0 & -2 & 3 \end{vmatrix}$$

$$\begin{vmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 4 & -2 \\ -1 & 0 & -2 & 3 \end{vmatrix} \times \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \\ -1 \\ 0 \end{vmatrix}$$

$x_3 = 0 \quad 2x_1 - x_2 - x_3 = 1 \quad -x_1 + 3x_2 = 0 \quad -x_1 + 3x_4 = 0$

$Ax = \begin{vmatrix} -1/2 \\ -1/4 \\ -1/4 \\ -1/2 \end{vmatrix} \quad CAx = \begin{vmatrix} -1/2 \\ -1/2 \\ -1/2 \\ -1/2 \end{vmatrix} \quad y = \begin{vmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{vmatrix}$ ✓

2.13

Review for exam!

Q1 u, v, w in R^3 they span a subspace in $R^?$ what are the possible Dimensions?

they can span $R^{0,1,2,3}$ max 3 because there is 3 vectors, non zero because its non-zero vectors ✓

Q2 5x3 u, $r = 3$

2.1 whats the null space? its the zero vector because all columns are pivots. ✓

2.2 $B = \begin{vmatrix} u \\ 2u \end{vmatrix}$ whats the rank? 3 ✓ whats the row echelon form? $\begin{vmatrix} u \\ 0 \end{vmatrix}$ ✓

2.3 $C = \begin{vmatrix} u & u \\ u & 0 \end{vmatrix}$ whats the reduced row echelon form? $\begin{vmatrix} u & 0 \\ 0 & u \end{vmatrix}$ ✓ whats the row echelon form? $\begin{vmatrix} u & u \\ 0 & -u \end{vmatrix}$ ✓ whats the rank of C? 6 ✓

2.4 whats the $\dim N(C^T)$ (C is $\overset{m}{10} \times \overset{n}{6}$) $= m - r = 4$ ✓

Q3 $Ax = \begin{vmatrix} 2 \\ 4 \\ 2 \end{vmatrix} \quad x = \begin{vmatrix} 2 \\ 0 \\ 0 \end{vmatrix} + c \begin{vmatrix} 1 \\ 1 \\ 0 \end{vmatrix} + d \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix}$ (3x3) whats the dimension of the row space? 1

3.1 $\dim N(A) = 2$ ✓

3.2 whats the matrix? $A = \begin{vmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 1 & -1 & 0 \end{vmatrix}$

3.3 $Ax = b$ can be solved if? if b has the form of $\begin{vmatrix} x \\ 2x \\ x \end{vmatrix}$ ✓

Q4 True or False

i) the null space of A is only the zero vector and A is a square matrix does that apply to A^T? T ✓

5x5 matrices as a vector space, do the invertible ones form a subspace? F ✓

if $Ab=0$ then $b=0$? T ✗ example $\begin{bmatrix}0&1\\0&0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$

nxn matrix, indeps col, $Ax=b$ always solvable? T ✓

if $m=n$ then the $C(A^T)=C(A)$. F ✓

the matrix A and $-A$ share the same 4 subspaces. T ✓

if A and B have the same 4 subspaces then A is a multiple of B. T ✗ can be debunked by an independent 6x6 matrix a and b

Q5 $B=\begin{bmatrix}1&1&0\\0&1&0\\1&0&1\end{bmatrix}\begin{bmatrix}1&0&-1&2\\0&1&-1&1\\0&0&0&0\end{bmatrix}$ without multiplying it. note: for an invertable C, $N(CD)=N(D)$

3x4

basis for $N(CD)=\begin{bmatrix}-1\\1\\0\\1\end{bmatrix}\begin{bmatrix}-2\\0\\1\\0\end{bmatrix}$ ✓

Solve $Bx=\begin{bmatrix}0\\0\\0\end{bmatrix}$. $x_p+x_n=\begin{bmatrix}1\\0\\0\end{bmatrix}+c\begin{bmatrix}1\\1\\0\end{bmatrix}+d\begin{bmatrix}2\\0\\1\end{bmatrix}$

Q6 A=nxm matrix

if i exchang 2 rows of A which subspaces stay the same? Row space, left hand null space, null space ✓

$V=\begin{bmatrix}1\\2\\3\end{bmatrix}$ can be in nullspace of a matrix and be a row, why not? if it was a row, then the nullspace would be $\begin{bmatrix}5\\14\\5\end{bmatrix}$